

CS 217: Practice questions on Linear regression and classification

- Suppose you have a regression task of predicting a real value in the interval $[-4, 4]$ on a d -dimensional input $\mathbf{x} = [x_1, \dots, x_d]$
 - First, assume we fit a model $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0$ on a training dataset $D = \{(\mathbf{x}^1, y^1), \dots, (\mathbf{x}^N, y^N)\}$ by minimizing the square loss. Let's say we know the optimal value of the $\mathbf{w}$ parameters. Call it $\mathbf{w}^*$. We only need to fit w_0 using the training dataset so as to minimize this loss. Provide a closed form expression for the fitted w_0 ? $w_0 = \frac{(y^i - \mathbf{w}^* \mathbf{x}^i)}{N}$..1
 - In the above if we change the loss to $\max(0, (y^i - f(\mathbf{x}^i))^2 - 1)$, and use gradient descent algorithm to find the optimal w_0 , write the updated w_0 if initial $w_0 = 0$ and learning rate is η . $-\eta \sum_{i=1}^N \delta((y_i - \mathbf{w}^* \mathbf{x}^i)^2 \geq 1)(y^i - \mathbf{w}^* \mathbf{x}^i)/N$..2
 - Instead of using a linear regression function with the least square training objective, you convert this into a four class classification task as follows. First discretize $[-4, 4]$ into these four intervals

$$[-4, -2), [-2, 0), [0, 2), [2, 4] \quad (1)$$

Call each of these as class 1, 2, 3, 4 respectively. In the training data, any example with y_i satisfying $-4 \leq y_i < -2$ will be assigned class 1, any with y_i satisfying $-2 \leq y_i < 0$ will be assigned class 2, and so on. When the predicted class is "2" by the classifier, the predicted y is midpoint "-1" of the corresponding interval $[-2, 0)$.

Show a dataset where you expect the linear regression method to be better than the above classifier method, and another where the opposite is better. 1-dimensional dataset suffices. *If original training data fits nicely with a straight line, then the classifier will incur discretization errors.*

If original data has clusters of piecewise constant regions than the classifier approach is better. ..4

- Let $\mathbf{w}^y$ denote the parameters associated with class y . A limitation of training the classifier model with the default softmax loss below

$$\min_{\mathbf{w}^1, \mathbf{w}^2, \mathbf{w}^3, \mathbf{w}^4} \sum_{i=1}^N \mathbf{w}^{y_i} \mathbf{x}^i - \log \sum_y e^{\mathbf{w}^y \mathbf{x}^i}$$

is that the error function does not recognize that the four class labels are ordered. Ideally, the loss of misclassifying an example to a near-by class should be smaller than to a far-off class. Suggest a modification to the above loss function to fix this limitation. *There are many possible ways to fix this. Here is a simple idea. When correct class is k , and predicted class is j , we assign a cost c_{jk} that is equal to the distance between the intervals of the two classes. Then, we minimize the following error:*

$$\min_{\mathbf{w}^1, \mathbf{w}^2, \mathbf{w}^3, \mathbf{w}^4} \sum_{i=1}^N \sum_y c_{y y_i} (\mathbf{w}^{y_i} \mathbf{x}^i - \log(e^{\mathbf{w}^y \mathbf{x}^i} + e^{\mathbf{w}^{y_i} \mathbf{x}^i}))$$

- Very often the training data for a classification task may assign wrong labels y for an instance. One idea to fix the above problem is to make the loss function noise tolerant. Let $P_\theta(y|\mathbf{x})$ be the probability of some outcome that we want to maximize. The standard cross entropy loss minimizes the loss $L(\theta) = -\log P_\theta(y|\mathbf{x})$.

- (a) Plot $L(\theta)$ as a function of $P_\theta(y|\mathbf{x})$ on the x-axis. ..1
- (b) Comment on why the above loss function is not a good idea when the label y is noisy. ..2
- (c) Suppose we modify the above loss to a generalized cross entropy loss defined as follows:
 $L_G(\theta) = -\log(P_\theta(y|\mathbf{x}) + 0.1)$. Plot $L_G(\theta)$ as a function of $P_\theta(y|\mathbf{x})$ on the x-axis. ..2

Simple

3. Suppose we wish to train a square error regression function with a L_γ regularizer. That is the training objective is made: $\min_{\mathbf{w}} \sum_i (y^i - \mathbf{w} \cdot \mathbf{x}^i)^2 + C \|\mathbf{w}\|^\gamma$ where $\|\mathbf{w}\|^\gamma = \sum_j |w_j|^\gamma$. Suppose, we decide to use co-ordinate descent as follows:

$t = 0, \mathbf{w}^t = 0$

while change **do**

for $j = 1 \dots d$ **do**

 /*Minimize wrt w_j keeping all other variables fixed at $\mathbf{w}^t$ */

$w_j = \min_{w_j: w_k = w_k^t \forall k \neq j} \sum_i (y^i - \mathbf{w} \cdot \mathbf{x}^i)^2 + C \|\mathbf{w}\|^\gamma$

$\mathbf{w}^{t+1} = (w_1^t, \dots, w_{j-1}^t, w_j, w_{j+1}^t, \dots, w_d^t)^T$

$t = t + 1$

end for

end while

- (a) Solve for the optimum w_j in step 4 for the case where γ is even and > 0 . ..3

Gradient: $2 \sum_i (y^i - \mathbf{w} \cdot \mathbf{x}^i)(-x_j^i) + 2Cw_j^{\gamma-1}$

$= 2 \sum_i (y^i - \mathbf{w}^t \cdot \mathbf{x}^i)(-x_j^i) + (w_j - w_j^t)(x_j^i)^2 + 2Cw_j^{\gamma-1}$

Equating the gradient to zero we get. $w_j(Cw_j^{\gamma-2} + \sum_i (x_j^i)^2) = \sum_i ((y^i - \mathbf{w}^t \cdot \mathbf{x}^i)(x_j^i) - w_j^t(x_j^i)^2)$

The closed form solution of this will depend on γ . For $\gamma = 2$ we get $w_j = \sum_i ((y^i - \mathbf{w}^t \cdot \mathbf{x}^i)(x_j^i) - w_j^t(x_j^i)^2) / (C + \sum_i (x_j^i)^2)$

- (b) Assume that for every C there exists a C' such that the training objective is equivalent to

$$\begin{aligned} \min_{\mathbf{w}} \sum_i (y^i - \mathbf{w} \cdot \mathbf{x}^i)^2 \\ \text{s.t. } \|\mathbf{w}\|^\gamma < C' \end{aligned} \quad (2)$$

Use this to write the optimal value w_j in terms of C' and $\mathbf{w}^t$ -s when $\gamma = 1$. ..4

Let $\alpha = C' - \sum_{k \neq j} |w_k|$

We need to minimize the objective while ensuring that $|w_j| < \alpha$. Using the solution from the previous question without the regularize we obtain $w_j = \sum_i ((y^i - \mathbf{w}^t \cdot \mathbf{x}^i)(x_j^i) - w_j^t(x_j^i)^2) / (\sum_i (x_j^i)^2)$ If w_j does not satisfy the condition $|w_j| < \alpha$ we shrink it to the nearest value that does satisfy the condition.

4. Consider the logistic classifier. If two features f_k and f_j are replicas of each other, state with a brief reason how the parameters w_k and w_j are related in the following two versions of the objectives used for training logistic classifiers: Assume the training data consists of instances of the form $(\mathbf{x}^i, y_i)$ where $y_i \in \{-1, 1\}$

- (a) Unregularized: minimize $\sum_i \log(1 + e^{-\mathbf{w}^T \mathbf{x}^i y_i})$..1 There is no unique solution. All possible values of w_j and w_k whose sum is a constant γ will give rise to the same optimum value.
- (b) Regularized with L2: minimize $C \sum_i \log(1 + e^{-\mathbf{w}^T \mathbf{x}^i y_i}) + \sum_j w_j^2$ where C is a constant. ..2 In this case w_j and w_k will be equal because for a fixed value of the sum, the L2 regularizer is minimized when the two parameter values are equal.
- (c) Regularized with L1: minimize $C \sum_i \log(1 + e^{-\mathbf{w}^T \mathbf{x}^i y_i}) + \sum_j |w_j|$ where C is a constant. ..2 In this case also there is no unique solution. All possible values of w_j and w_k such w_j and w_k have the same sign but whose sum is a constant will give the same objective value.

5. In class we only considered square error for training a regression model. Propose an alternative error function and discuss pros and cons on account of accuracy, sensitivity to noisy training data, and training time.

Option 1: Absolute loss: $|y^i - \mathbf{w}^T \mathbf{x}^i|$

This loss function is better behaved than the Square loss because it is less sensitive to outliers in the training data. The training time will be higher because the function to be optimized is not differentiable. However, the training objective with the above loss is convex and can be solved using constrained convex optimization techniques.

Option 2: Clipped loss: $\min(|y^i - \mathbf{w}^T \mathbf{x}^i|, T)$ where T is a user specified threshold.

This is even more suited to handle outliers in training data because the loss will not continue to increase as we clip it at T . However, the training objective in this case, is neither differentiable nor convex. Therefore, it is difficult to guarantee a globally optimal value for $\mathbf{w}$. ..4

6. Let's say that we are trying to fit the parameters of a 3-way multinomial distribution using maximum likelihood estimation principle. We express the probability of the multinomial in terms of parameters $\{l_1, l_2, l_3\}$ as follows

$$f(x, \mathbf{w} = \{l_1, l_2, l_3\}) = \frac{e^{l_x}}{e^{l_1} + e^{l_2} + e^{l_3}}$$

For example, if $l_1 = 0, l_2 = 1.4, l_3 = 2$, then the probability of $x = 2$ is $\frac{e^{1.4}}{1 + e^{1.4} + e^2}$

- (a) Using the above form, write the log likelihood of the dataset $D = \{x^1, \dots, x^N\}$ where each x^i is either 1, 2, or 3.

$$\sum_{i=1}^N l_{x^i} - N \log(e^{l_1} + e^{l_2} + e^{l_3})$$

..1

- (b) Write the gradient of the above objective with respect to parameter l_j for a training instance x^i .

$$\delta(x^i == j) - \frac{e^{l_j}}{(e^{l_1} + e^{l_2} + e^{l_3})}$$

..1

- (c) If the initial value of the three parameters is $l_1 = 0, l_2 = 2, l_3 = -1$, for which value of x^i is the norm of the gradient highest? Justify your answer. Corresponding to the j for which the initial multinomial probability is the smallest, that is, 3. ..2
- (d) If we obtain MLE estimate for the above model using the stochastic gradient descent algorithm with batch size one on a data set where outcome 3 is never observed, what is the smallest possible value of l_3 at the end of 100 gradient updates with a learning rate η and initial value of $l_1 = l_2 = l_3 = 0$ This is an open-ended question, and any decent lower bound on the value l_3 after 100 iterations would be accepted. Here is a loose bound. The gradient wrt l_3 at any iteration t is $-\frac{e^{l_3^t}}{(e^{l_1^t} + e^{l_2^t} + e^{l_3^t})}$. Initially, the value of the gradient is $1/3$. Since 3 never appears in the dataset, in subsequent iterations, while l_1, l_2 could increase, l_3 keeps on decreasing. Thus the value of the gradient of l_3 is never greater than $-1/3$. This gives us a lower bound on the value of l_3 after 100 iterations to be $-100\eta(\frac{1}{3})$. ..3
- (e) Now say we redefine the probability using a smoothing constant ϵ as

$$f_\epsilon(x, \mathbf{w} = \{l_1, l_2, l_3\}) = (1 - 3\epsilon) \frac{e^{l_x}}{e^{l_1} + e^{l_2} + e^{l_3}} + \epsilon$$

Write the gradient of the maximum likelihood estimation objective of the above probability distribution with respect to parameter l_2 for a training example $x^i = 2$. We need to take the gradient of $\log(\epsilon + (1 - 3\epsilon) \frac{e^{l_2}}{e^{l_1} + e^{l_2} + e^{l_3}})$ wrt l_2 ..3

- (f) What are the pros and cons of using f_ϵ Vs f for estimating density of a K way categorical variable?

..2

7. Suppose we train a model for classification using the same square loss that is used for training regression models. Assume the labels are either +1 or -1. And our training objective is:

$$\min_{\mathbf{w}, w_0} \sum_{i=1}^N (y_i - (\mathbf{w}^T \mathbf{x} + w_0))^2 \quad (3)$$

Assume that the training data is separable by a linear decision boundary. Is the classifier trained using the above objective guaranteed to achieve zero error on the training data? Support your answer with proper justification. ..1 No. Let $d = 1$, there are many negative instances with $x_1 = -1, y = -1$, many positive instances with $x_1 = 1, y = 1$ and a single positive instance with $x_1 = -0.1, y = 1$. The least square minimizer is $w_1 = 1, w_0 = 0$ which does not correctly classify the last positive instance at $x_1 = -0.1$.

8. Let $f(x) = \max(0, 1 - x)$ Find the subgradients of the function at 0, 1, and 2?
9. Questions 11.1 to 11.5 in Chapter 11.8 of PLS textbook.